BETAMETHASONE

Antenatal lung maturity

Potent topical steroid

CHEMICAL NAME 9-fluoro-11β,17,21-trihydroxy-16β-methylpregna-1,4-diene-3,20-dione (sodium phosphate / acetate / valerate / dipropionate)

CLASS Long-acting potent corticosteroid (glucocorticoid)

TRADE NAMES Celestone Soluspan · Betnesol · Betnovate (topical) · Diprosone

THERAPEUTIC USES / INDICATIONS

- Antenatal corticosteroids for fetal lung maturation (24–34 weeks).
- Topical: eczema, psoriasis, dermatitis (potent).
- Systemic inflammatory/allergic disorders; cerebral oedema.
- Eye, ear and nasal drops; intra-articular injection.

ADMINISTRATION

- **IM** (deep) for antenatal course.
- **Topical:** thin layer once or twice daily (fingertip units).
- **PO** soluble tablets; drops.

DOSAGES

- **Antenatal:** 12 mg IM, repeated after 24 h (2 doses).
- **Topical (0.1% valerate):** apply 1–2 times daily; short courses on face.
- **Systemic:** 0.5–5 mg/day.

PHARMACOKINETICS

- **Duration:** 36–54 h (long-acting).
- Metabolised in liver.
- Excreted in urine. **t½:** 6.5 h.

PHARMACODYNAMICS

- Potent **glucocorticoid** (~25× hydrocortisone); negligible mineralocorticoid effect.
- Crosses placenta → fetal surfactant production, ↓ RDS, IVH, NEC.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Topical: skin thinning, striae, telangiectasia, acne, perioral dermatitis.
- Systemic: hyperglycaemia, insomnia.

ADVERSE EFFECTS (SERIOUS)

- Adrenal suppression (systemic or extensive potent topical use).
- Neonatal hypoglycaemia (antenatal).
- Infection, osteoporosis (long-term).

CONTRAINDICATIONS & CAUTIONS

- Topical: untreated skin infection, rosacea, acne, perioral dermatitis.
- Systemic infection without cover.
- **Caution:** diabetes (maternal glucose rises).

KEY INTERACTIONS

- **Antidiabetics:** ↑ glucose.
- **Enzyme inducers:** ↓ effect.

NURSING MANAGEMENT

- Antenatal: record dose times; monitor maternal glucose (especially diabetics) for 72 h.
- Topical: teach fingertip units; avoid face and skin folds unless prescribed.
- Inform neonatal team of steroid course timing.

PATIENT EDUCATION

- Use creams sparingly and only for the time advised.
- Do not use on face unless instructed.
- Antenatal: explains the course helps baby's lungs.

KEY POINTS

- Long-acting potent glucocorticoid.
- Potent topical steroid — skin atrophy.
- Antenatal: 12 mg IM × 2 doses 24 h apart.
- Maternal hyperglycaemia.